

Large Language Model (LLM)-powered Autonomous Agents and Multiagent systems

- 11:59 pm 02/11/2025 (End of Week 12)
- **Contributes** 50% of your final result
- **Group Assignment – Group of 3-5 students**

To do this Option D, you will need to apply with the unit convenor by presenting your suitability to do this research-oriented project: Your current average from your academic transcript has to be at least D ($\geq 70\%$) AND you are aiming to achieve D/HD in Intelligent Systems AND you are well equipped to do this project by having a strong capability to do Python programming.

Summary

This project requires you to work in a team led by a teaching staff aiming to implement and demonstrate the capabilities of autonomous agents powered by large language models (LLMs) such as GPTs, Llama's, or Gemini's. A simple introduction to this concept can be found here: <https://botpress.com/blog/llm-agents>. (Students are expected to do own research to identify relevant and useful learning resources.)

There are two main goals in this project:

1. **(G1)** Develop and deploy an LLM-powered autonomous agent; and
2. **(G2)** Develop and deploy a multiagent system consisting of multiple interacting LLM-powered agents.

While we would ideally like to achieve both goals, if the team encountered significant challenges, then we would aim to achieve at least **(G1)** and try to get as much done toward **(G2)** as the team can.

One of the important things to keep in mind when building an agent is the purpose of the agents, i.e., what an agent (or, a multiagent system) is supposed to do. While there are many interesting applications (e.g., a personal assistant agent, a network of sales agents, IT support agents, etc.), we will focus on building agents and/or multiagent systems whose mission is one of the following:

1. **Business improvement agent:** The agent accesses the reviews about a company or organisation and provides a summary of the customers' general sentiments as well as identifiers areas for business improvement.
2. **Cybersecurity support agent:** The agent retrieves system logs to diagnose issues, detect (and potentially predict) cyber attacks and combine intelligence from CTI (Cyber Threat Intelligence) reports to help the organisation in proactive defence, incident response, and improving overall security posture.

Project requirements

- The project will be done in Python. If you have zero or little experience with Python, this project is not suitable for you.
- Source code maintained on Git based VCS (Github/Bitbucket/GitLab/...). You must provide read-only access to the teaching staff supervising your project.
- Project documentation maintained on OneDrive. You must provide access to the teaching staff supervising your project.
- Learning about LLMs, agentic AI, autonomous LLM agents, LLM-powered multiagent systems, etc.
- Implementing and deploying a simple LLM agent by following some available tutorials (we expect that every student will be able to implement the agent with locally-deployed LLMs and the team will also demonstrate LLM agents using LLMs deployed on cloud).

- Implementing and deploying LLM agents whose mission is agreed between the team and the supervisor.
- Implementing and deploying a LLM-powered multiagent system that consists of multiple agents with different capabilities who collaborate to achieve the team's common mission.
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- Implementing a UI for the user to interact with the developed system(s).
- Running illustrative demo of a working prototype. Ideally, this prototype should be run on a server for the user to access the service from their browser (similar to ChatGPT).
- Evaluation of the system/model against benchmark evaluations/datasets.
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- Portfolio consisting of:
 - Individual **weekly worklogs**;
 - Individual **fortnightly reports on the work done/achievements**;
 - A team **project summary report**.
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- The **project summary report** (20-30 pages) includes the following sections:
 - Cover Page (with team details) and a Table of Contents (TOC),
 - Introduction,
 - Overall system architecture (including details of the external libraries or software packages used by your solution),
 - Detailed installation/user guides,
 - Details of the implementations members of the team have done,
 - Details of the experimentations and evaluation members of the team have carried out,
 - Scenarios/examples to demonstrate how the system works,
 - Some critical analysis of the implementation,
 - Analysis/Discussion of the experimental results, and
 - Summary/Conclusion.
- A video presentation of the project (12-15 minutes)

Option D – Marking Scheme:

Each member of the team will be assigned weekly tasks. Depending on the effort put into the individual tasks + the quality of the work, the individual will receive fortnightly marks between **0 and 14 for the five fortnights (~ ten weeks 3-12)** and feedback will be given during weekly team meetings. Each individual's final result for the Project will be the sum of the individual's marks for the ten weeks 3-12 plus the team portfolio assessment.

The team portfolio (maximum **30** marks) will be assessed as follows:

- Project report: Up to **20** marks
- Video presentation: Up to **10** marks
- Individual contributions: Up to **-30** marks (that is, even if the team portfolio is awarded 30 marks, a student who makes no/little contributions may get 0 mark for this component)

Submission

At least one member of the team must submit the entire project (code + report) as a .zip file to Canvas by 11:59pm of 02/11/2025. Create a single zip file with your code and a working version of your system. Standard late penalties apply – 10% for each day late, more than 5 days late is 0%.

You must also provide your project supervisor read-only access to your git repository within 1 week of forming teams.

The video (**12-15 minute duration**) should be submitted to Canvas by at least one member of the team by 11:59pm on Tuesday 04/11/2025.